

Experiment 1B

Aim

File Sharing Within an Organization Using IP Address

Theory

In a local area network (LAN), computers communicate with one another using unique numeric identifiers called **IP (Internet Protocol) addresses**. File sharing within an organization relies on the **Server Message Block (SMB)** protocol over TCP/IP, which allows client applications to read and write to files on a host server.

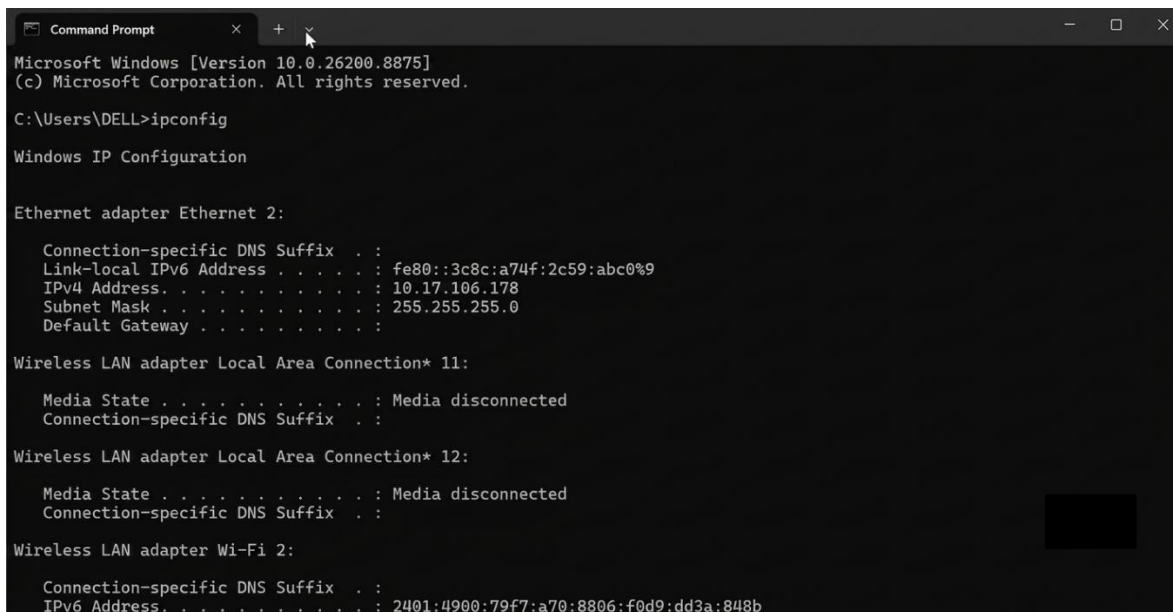
Steps Executed in Theory:

1. **Host Network Identification:** The host machine's IP address and network adapter details are retrieved using the network configuration command:

Ipconfig

2. **Network Connectivity Verification:** The connection stability and network latency between workstations are verified using the Internet Control Message Protocol (ICMP) echo request tool:

ping 10.18.116.158



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::3c8c:a74f:2c59:abc0%9
    IPv4 Address. . . . . : 10.17.106.178
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 11:

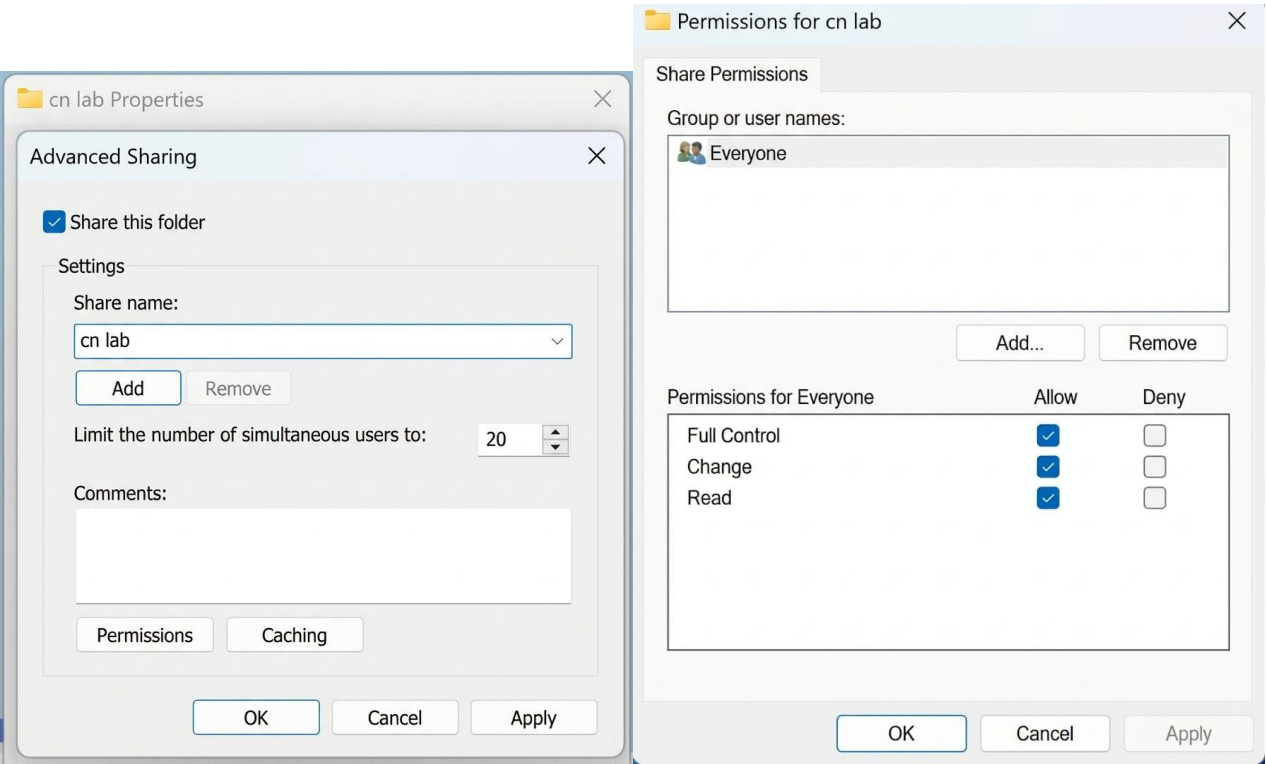
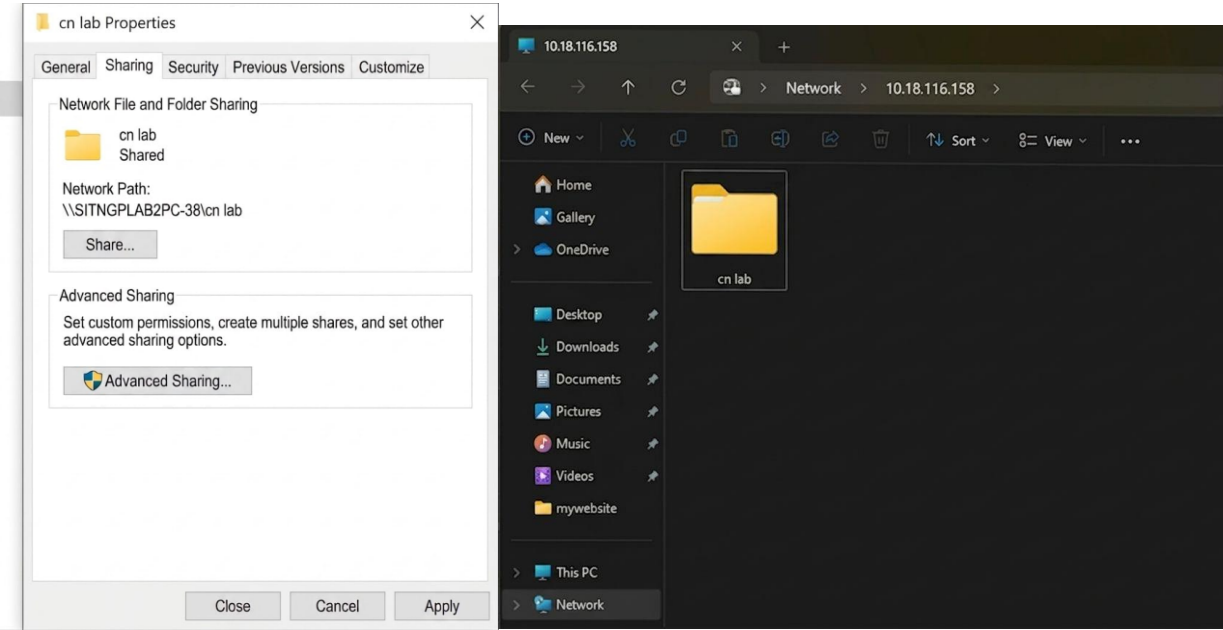
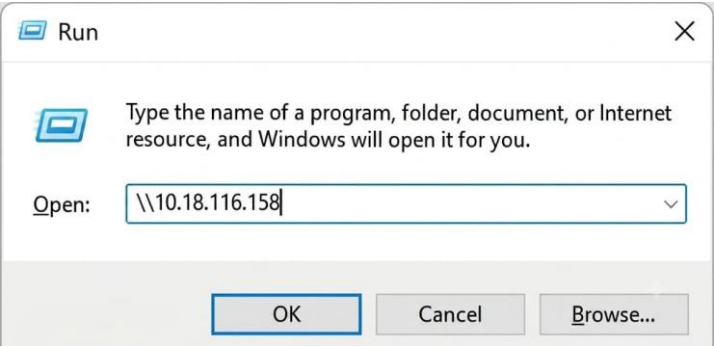
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 12:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 2:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2401:4900:79f7:a70:8806:f0d9:dd3a:848b
```



3. **Directory Sharing Configuration:** A target directory is configured on the host machine (sit.lab2) with active read/write permissions enabled under **Properties -> Sharing -> Advanced Sharing**.
4. **Remote Access via Universal Naming Convention (UNC):** Client machines on the same subnet (255.255.255.0) connect directly to the shared resource over TCP/IP using the syntax:

\\10.18.116.158\Shared_Folder

Conclusion

The network interface configurations were successfully retrieved using the ipconfig command, and IP connectivity was verified using the ping utility with a 0% packet loss rate. File sharing within the organization was established and executed using the host system's IP address (\\10.18.116.158).